

Review

Presence Assessment in Virtual Reality: A Systematic Literature Review

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Abstract

A critical aspect of virtual reality is the extent to which the user forgets their real surroundings and becomes completely engaged within the virtual environment. Diverse factors affect this user perception, which are grouped into two main concepts: immersion and presence. Although the study of presence is extensive, researchers have not reached a consensus on a protocol with specific instruments and stages to evaluate it. This leads to a wide variety of results with different assessment methods, experimental setups, stimuli implemented, and applications. Therefore, this article aims to provide an analysis of the state-of-the-art methods for assessing presence in VR systems during the last few years. This study seeks to determine and improve the understanding of current techniques used for presence assessment, human data collected, data analysis methods, and the technologies and virtual environments implemented. In addition, four opportunities are discussed to provide researchers guidelines that can lead to enhanced presence assessments and personalized VR experiences.

Keywords: presence assessment; virtual reality; presence; immersion

1. Introduction

Research on virtual reality (VR) is extensive and spans a wide range of applications. For instance, in applications for training and simulation, VR has been used in aviation [1,2], driving tasks [3–5], training performance [6–8], social conscience [9], and firefighting [10]. In the education sector, several activities and routines can be enhanced by using VR [11–13], offering more innovative and interactive solutions in classrooms. In healthcare, VR has shown positive results in pain management and physical therapy, providing patients with an alternative to traditional, often monotonous routines [14–17]. These studies (and many more) demonstrate the importance of such technology across a wide range of human activities. Nevertheless, even today, it is essential to ask: What makes VR a promising technology to explore in such diverse fields? First, VR enables the emulation of real-life situations and artificial scenarios in a controlled, safe environment. Within this framework, activities that, in some cases, could pose a risk to those who carry them out (e.g., treatment of phobias, learning to drive or fly) can be implemented, allowing for the exploration of human behavior without compromising its integrity [18]. Moreover, emulation in VR environments helps practitioners and researchers test prototypes or experimental setups,



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substantially reducing costs and time during assessment. A notable example is digital twins, which, when enabled by VR environments, deliver innovative results in agriculture [19], smart cities [20], and robotics [21].

Regardless of the application, a critical issue in VR research is how users perceive virtual environments as real, and to what extent they forget their real surroundings and become fully engaged in the virtual environment. Numerous factors affect each user's perception, but VR research has focused on two fundamental concepts: immersion and presence. In the past, both concepts were often confused or used interchangeably (erroneously); however, today, researchers identify immersion as the objective level of sensory fidelity provided by the VR system. In other words, it relates to the technology implemented in the VR system and the extent to which the system can provide information to the user in a vivid way that stimulates the primary sensory systems [22,23]. Presence is considered a mental and physical state in which users feel as if they are in a place when they are not. It has been defined as the "human response to immersion" [24], which constitutes a complex state that is generated individually through the processing and interpretation of virtual reality stimuli.

It has been suggested that the more feedback elements are used in a virtual reality system (VRS), the more immersive the system is and the greater the presence achieved [25,26]. Over the years, researchers have developed enhanced immersive VR systems by implementing olfactory [27–30] and thermal–sensory feedback [31,32]. In addition, systems such as treadmills for virtual climbing [33] or haptic vests [32] represent novel approaches to provide feedback and increase immersion. Nevertheless, some authors have discussed the real impact of high-fidelity feedback systems on user presence, raising the question: Does a more immersive system always ensure a high state of presence in users? Several years ago, Bowman et al. discussed the possibility of achieving the presence state with minimal immersive elements [23]. It was stated that not all applications require sophisticated immersive elements, so it is unnecessary to incur expenses on such systems. In addition, some insights on virtual environments and the role of fidelity in immersion were evaluated by [34–36]. As expected, while greater graphics detail and improved display rendering enhance the user experience, they do not yield statistically significant improvements. The authors note the importance of user interactive tasks, arguing that an engaging task can provide greater immersion than improved graphics. Weber et al. [37] summarize part of these findings, noting that a system that combines several feedback elements with high display capabilities (a highly immersive system) does not guarantee high presence among users.

Due to the uncertain effects of immersive hardware and its ongoing evolution, most of the literature focuses on presence assessment. The main goal of understanding presence in virtual reality is to provide researchers and practitioners with the knowledge and tools to deliver better virtual experiences for users. To achieve this, not only must the elements of the VR system be considered, but also diverse human factors, affective states, performance metrics, and even how to avoid cybersickness [35,38]. Although the study of presence is extensive, there is no consensus on a protocol with specific instruments and stages for its evaluation, resulting in a wide variety of results that can be challenging and time-consuming for researchers to interpret. The main inconsistencies found in the literature are as follows:

- Some studies conduct the presence assessment solely based on subjective measures (questionnaires), while others prefer a combined approach with affect detection.
- The affect detection strategies are often focused on a limited set of physiological signals.

- There are no standards for designing and implementing VR stimuli to evoke presence; VR environments based on commercial video games, 360° videos, and in-house developments can be found; the latter are most recommended, as they provide full control of the environment.
- There is a lack of control approaches to modify VR stimuli to enhance VR experiences and enable users to achieve presence.

Therefore, this literature review aims to provide an analysis of the state of the art in the assessment of presence in VR systems over the last few years and to discuss future directions to advance the state of the art. The main goal of this article is to determine and improve understanding of the current techniques used for presence assessment, the human data collected, the technologies and virtual environments implemented, and to discuss the opportunities that can pave the way for enhanced presence assessments and personalized VR experiences. The development of this review does not imply that other VR review articles do not exist. However, such papers are often focused on specific domains of VR and do not address the points described above. For example, Refs. [39–41] present extensive literature reviews on the elicitation and recognition of affective states. They discuss affective models, VR elements that induce affective states, and affective computing analysis techniques. The survey that most closely aligns with this review of the literature is that presented by [42]. They elaborated on a broad literature review of the concept of presence and related concepts, such as plausibility, social presence, embodiment, models of presence, and the measurement of presence using questionnaires. Therefore, this review seeks to answer the following research questions:

- **RQ1.** What approaches and instruments are currently used to evaluate presence in VR?
- **RQ2.** What kind of VR stimuli and hardware have recently been used to evoke presence?
- **RQ3.** Given the link between presence and affective states, which physiological signals are utilized in the affective computing strategies?
- **RQ4.** What areas of opportunity exist regarding presence in VR, and what could be the next steps toward personalized experiences?

The rest of this paper is organized as follows. In Section 2, some concepts of presence and its essential associated elements are presented. Later, in Section 3, the search methodology for this review is explained, including a set of keywords and exclusion criteria used to select a specific set of articles. The synthesis of the literature highlighting the most important trends is developed in Section 4. Later, Section 5 discusses the main findings, while a set of future directions designed to help researchers further advance the state of the art is presented in Section 6. Finally, an overall conclusion is presented in Section 7.

2. Background

The study of presence in VR dates back to the 1990s. Since then, several definitions of presence have been proposed that, regardless of their publication year, present similar key ideas [37]. Table 1 shows different definitions of presence over the years. It can be observed that some definitions have received many citations over the years. It is important to take them into account and compare them with newer ones, regardless of their release year, so that new researchers can adopt the one that best fulfills the objective of their work. Moreover, it is also important to note that, according to experts in the VR community over the years, reaching the state of presence is “the central goal of virtual reality” [18,43,44]. Attempting to define a single definition of such a complex concept is challenging and controversial. Nonetheless, we consider that the presence assessment

process in VR scenarios involves the VR system providing artificial stimuli to the user to help them become immersed and temporarily forget their real surroundings. Despite the task (interactive or not), it is desirable to maintain users engaged within the VR dynamics to maximize a specific objective, such as learning, relaxation processes, or entertainment, as happens within the “flow state” in video games [45]. Ultimately, the concepts of the VR community are designed to be helpful for the design of VR experiences.

Table 1. Different conceptions about presence through the years.

Source	Citations	Year	Definition
[46]	1183	1992	The sense of being physically present with visual, auditory, or force displays generated by a computer.
[47]	5944	1997	The perceptual illusion of non-mediation.
[22]	3248	1997	A state of consciousness, the (psychological) sense of being in the virtual environment.
[43]	8800	2000	The experience of one’s physical environment; it refers not to one’s surroundings as they exist in the physical world but to the perception of those surroundings as mediated by automatic and controlled mental processes.
[48]	-	2000	A psychological state or subjective perception in which, even though part or all of an individual’s current experience is generated by and/or filtered through human-made technology, part or all of the individual’s perception fails to accurately acknowledge the role of the technology in the experience.
[49]	2371	2004	A psychological state in which virtual objects are experienced as actual objects in either sensory or non-sensory ways.
[50]	89	2013	A subjective experience of being bodily or physically located in a mediated environment.
[11]	124	2018	A psychological state of existence within an environment.
[51]	132	2019	The cognitive or psychological phenomena that a user is mediated and addicted in media technologies such as televisions, movies or games.

Given its complexity, presence depends on numerous human factors and hardware features. It would be impractical to try to evaluate or control each of these elements; however, researchers have identified key aspects to consider during VR development and presence assessment, regardless of the type of virtual environment, the hardware used, or the objectives of the VR system.

2.1. Presence’s Illusions

One of the most interesting aspects of virtual reality is its ability to generate illusions that are not possible with other media. In the literature, presence illusions are defined as perceptions generated by digital sources that are different from what is actually perceived [42,52–54]. That is, a person may observe some object that reflects light, casts shadows, appears to have a particular texture, but in reality, they are seeing only two small screens (one per eye) displaying illuminated pixels in a variety of colors. In the same way, the movement inside virtual environments is actually a continuous change in the color of the pixels. Ref. [52] identifies four types of illusions:

- **Place illusion:** This concept refers to the specific sense of “being there” and was proposed by [55] with the name “Telepresence” to describe the similar feeling that can arise when embodying a remote robotic device in a teleoperation system. In other words, place illusion is the sensation of feeling present within the virtual environment or as part of it [56,57].
- **Plausibility:** This illusion refers to the importance of generating virtual scenes whose dynamic behavior is congruent with the actual environment that the VR system is recreating. Generating plausibility represents an important challenge for developers because it requires guaranteeing the congruence and credibility of what happens in the virtual environment. For example, a flight simulator must fulfill specific hardware features and dynamics during operation while flying in a consistent virtual environment. Otherwise, the experience may be unrealistic for its target users.
- **Body ownership:** This illusion refers to the capability of a VR system to allow its users to see or feel their body through the avatar in the simulation. Through real-time body tracking, the virtual body should be able to move synchronously and correspondingly with its movements. Similarly, sensory feedback is desirable when the virtual body interacts with the virtual environment. This illusion is commonly also called embodiment.
- **Copresence:** Essentially, it refers to the degree to which a participant feels the illusion of being with other physically remote individuals in a VR system. In other words, it establishes a sense of virtual togetherness [58].

Depending on the virtual reality, it may be possible to implement neither body ownership nor copresence while still ensuring presence. These two illusions of presence help to generate and enhance the virtual reality experience, but their inclusion is not mandatory and depends on the application’s requirements.

2.2. Relationship Between Presence and Affective States

At the beginning of the 2000s, affective computing (AC) systems emerged as a novel approach to analyzing the autonomic nervous system (ANS) responses of users, which are instantaneous, involuntary reactions of the human body linked to their emotional states [59]. To carry out an affective state estimation, physiological cues like the electromyogram (EMG), the electrocardiogram (ECG), electrodermal activity (EDA), Photoplethysmography (PPG), and electroencephalographic (EEG) signals, are processed to obtain time and frequency domain features that correspond to specific patterns in subjects’ affective states [60].

When a user uses VR and enters a virtual environment with HMDs or cave systems, their body responds in real time to the simulated experience, manifesting changes in diverse peripheral signals and generating movements or interactions in response to the virtual environment. Such signals and body dynamics provide quantifiable and reproducible data that offer objective metrics for assessing presence.

According to empirical evidence from presence studies over the years, a VR system that can elicit affective responses in users similar to those triggered by real-world tasks is consequently evoking a high sense of presence in its users [61–63]. For instance, among the most notable results validating the link between affective states and presence are those that evaluate stress, fear, or anxiety [17,64,65].

3. Methodology and Data Collection

As mentioned before, research on VR and presence has increased over the years. Figure 1 shows the overall upward trend of research on presence. There was a specific decrease in publications in 2021. Despite a lack of official reports, this decrease may be associated with the health contingency caused by COVID-19, making it difficult to conduct experiments with participants. For this review, research papers focusing on presence assessment in virtual reality studies were examined. The search was carried out using the keywords “Virtual Reality”, “Presence”, “Sense”, AND “Assessment” in the title, abstract, or keywords with a year span 2018–2025. The keyword “Virtual Reality” helps exclude works that use the term “Presence” in other contexts, like political discussions [66], or as a mechanism affecting consumers’ purchase intentions in shopping tasks [67].

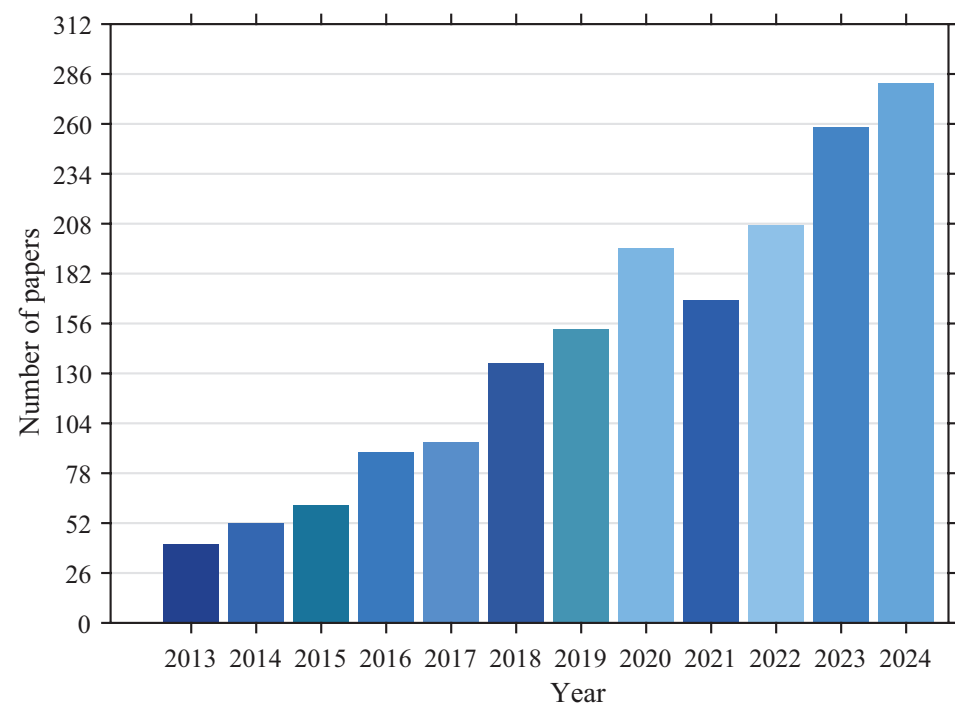


Figure 1. This graph reports the number of papers per year with the keywords “Virtual Reality” AND “Presence” AND “Sense” obtained from Scopus until December 2025.

A total of 1794 studies were identified in Scopus, along with 20 studies from Google Scholar. Later, a set of exclusion criteria following the PRISMA approach [68] was used to identify the most relevant works. As shown in Figure 2, 1704 articles were initially excluded according to three exclusion criteria. Some of those articles were unrelated to computer science, engineering, or psychology, and some were non-English manuscripts. Later, nine additional articles were excluded for not falling within the Q1 or Q2 quartiles according to Scimago’s impact factor scale, or for not having their respective registers. In total, 101 studies were used in this literature review.

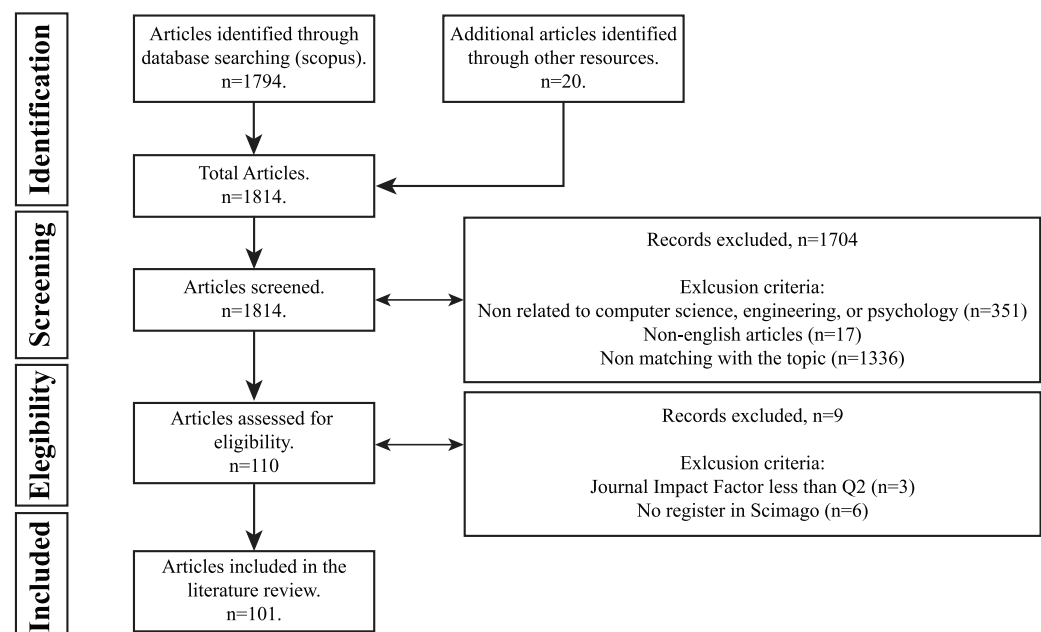


Figure 2. PRISMA approach for selecting the papers from 2018 to 2025.

4. Synthesis of the Literature Review

According to the articles selected using the PRISMA approach, the entire dataset was synthesized to identify the main presence-related trends. To present the information more clearly, the data were organized and categorized into sections that describe the main findings. Below are the data on presence factors, stimuli and sensory feedback, hardware and display devices, and presence assessment. These sections help address the first three research questions stated in Section 1. Table 2 synthesizes the reviewed literature, organized in ascending form, and where each column indicates a particular aspect of the trending factors involved in VR presence assessment. The reader will be referred to this table several times through the manuscript.

Table 2. Reviewed research.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Buttussi and Chittaro (2018) [69]	Aviation safety training	Presence, engagement	Questionnaires, statistical analysis	IPQ, FAS	None	Serious game simulating an aircraft emergency evacuation	Monitor, Sony HMZ-T3W, Oculus Rift DK2	Visual, auditory	Open loop
Zou et al. (2018) [70]	VR media quality evaluation	Spatial presence	Questionnaire, statistical modeling	5-point Spatial Presence Scale [71]	None	Omnidirectional (360°) video viewing of 8 video clips	HMD not specified, monitor	Visual, auditory	Open loop
Khan et al. (2018) [72]	Telepresence and remote collaboration	Spatial presence, social presence	Questionnaire, descriptive statistics	Custom presence questionnaire	None	Remote collaboration setup	Google Cardboard HMD	Visual, auditory	Open loop
García et al. (2018) [32]	Multisensory VR	Presence	Questionnaire, statistical analysis	Custom presence/realism questionnaire	None	Post-explosion train station scenario	HTC Vive	Visual, auditory, active and thermal haptic	Open loop
Waltemate et al. (2018) [73]	VR embodiment and avatar personalization	Presence, arousal, valence	Questionnaires, statistical analysis	Presence single-item rating, SAM	None	Simple virtual room with mirror-based avatar embodiment setup	HTC Vive, CAVE	Visual	Open loop
Gromer et al. (2019) [25]	VR exposure therapy for acrophobia	Fear, presence	Questionnaires, statistical analysis	AQ, STAI, SSQ, MEC-SPQ	ECG (HR), EDA (SCL)	High mountainous environment	HTC Vive	Visual, auditory	Open loop
Selzer et al. (2019) [74]	Educational VR	Presence	Questionnaires, statistical analysis	PQ, single-item presence questionnaire [75]	None	Virtual wetland of Villa del Mar (Argentina)	Monitor, VR-Box, Oculus Rift CV1	Visual, auditory	Open loop
Wei et al. (2019) [76]	Tourism	Presence	Questionnaire, statistical analysis	Presence scale	None	Real-world VR roller coaster	HMD not specified	Visual, auditory	Open loop
Jang and Park (2019) [51]	VR gaming	Presence and enjoyment	Questionnaires, structural equation modeling	Presence scale, enjoyment scale	None	Commercial VR gaming context	Not specified	Not applicable	Not applicable
Pallavicini et al. (2019) [77]	VR gaming, user experience	Happiness, surprise, anxiety, arousal, presence	Questionnaires, statistical analysis	Visual Analogue Scale, SUS	ECG (HR), HRV (LF/HF)	Smash Hit mobile videogame	Samsung Gear VR	Visual, auditory	Open loop
Shu et al. (2019) [78]	Disaster education	Spatial presence	Questionnaires, statistical analysis	TPI	None	3D earthquake simulation serious game	Monitor, HMD not specified	Visual, auditory	Open loop

Table 2. Cont.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Riches et al. (2019) [79]	Clinical psychology	Presence	Qualitative self-report	Semi-structured interviews	None	Virtual bar room	Oculus Rift DK2	Visual, auditory	Open loop
Chirico and Gaggioli (2019) [80]	Emotion research in VR	Awe, amusement, anger, disgust, fear, pride, sadness, joy, presence	Questionnaires, statistical analysis	PANAS, ITC-SOPI, custom Likert emotional ratings	None	360° nature environment	Samsung Gear VR	Visual, auditory	Open loop
Weech et al. (2020) [81]	VR user experience research	Presence	Questionnaires, statistical analysis	SSQ, PENS-PI, SIP, FMS	None	Lone Echo VR game	Oculus Rift CV1	Visual, auditory	Open loop
Cadet and Chainay (2020) [82]	Cognitive psychology and VR research	Presence, emotion (valence and arousal)	Questionnaires, statistical analysis	ITC-SOPI, SAM, PANAS	None	Island (wildlife scenes) and City (urban scenes), each with emotional 3D stimuli (positive, neutral, negative)	HTC Vive, monitor	Visual, auditory	Open loop
Bourhim and Cherkaoui (2020) [7]	Emergency training and human behavior simulation	Presence, anxiety, stress	Questionnaire, behavioral analysis, statistical comparison	Custom VR experience questionnaire, Anxiety/stress response questions (self-reported)	None	Residential building fire simulation	HTC Vive	Visual, auditory	Open loop
Jeong and Oh (2020) [83]	VR content evaluation and user experience research	Presence	Questionnaires, correlation analysis	PQ, ITQ	None	VR contents with different purposes	HMD not specified	Visual, auditory	Open loop
Athif et al. (2020) [84]	Presence evaluation in VR	Presence	Questionnaires	PQ, SUS	EEG, ECG (HR, HRV LF/HF), EDA (SCL, SCR)	Forest	Oculus Rift DK2	Visual, auditory	Open loop
Caldas et al. (2020) [85]	Serious games and emotional engagement	Valence, arousal, dominance, presence	Self-report emotional scales, behavioral performance analysis	SAM, VRSQ	ECG (HR, HRV LF/HF, RMSSD), EDA (SCL, SCR), RSP (breathing rate, breath variability)	Skydiving VR serious game	Oculus Rift	Visual, auditory	Open loop

Table 2. Cont.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Venkatakrishnan et al. (2020) [86]	VR user experience research	Presence	Questionnaires, structural equation modeling (SEM)	SSQ, SUS, NASA-TLX	EDA (SCL)	Virtual city driving simulation	HTC Vive Pro	Visual, auditory	Open loop
Gonçalves et al. (2020) [27]	Multisensory VR	Presence	Questionnaires, statistical analysis	IPQ, SSQ	None	Custom VR game “Illusions”	HTC Vive	Visual, auditory, passive haptic, olfactory	Open loop
Marto et al. (2020) [28]	Cultural heritage	Presence, enjoyment	Questionnaire, statistical analysis	IPQ	None	Roman archaeological ruins	Smartphone-based HMD	Visual, auditory, olfactory	Open loop
Hasanzadeh et al. (2020) [87]	Construction safety training and risk behavior analysis	Presence, arousal	Questionnaire, behavioral risk metrics	Single-item presence questionnaire [75]	HR, HRV	Rooftop construction hazard scenario	CAVE	Visual, auditory, passive haptic	Open loop
Khenak et al. (2020) [88]	Teleoperation and navigation performance research	Spatial presence	Questionnaires, behavioral metrics, statistical analysis	SPIE, SSQ	None	Warehouse-style navigation task	HTC Vive Pro	Visual, auditory	Open loop
Uhm et al. (2020) [89]	Sport marketing	Arousal, presence	Questionnaire, statistical analysis	Presence scale	EEG	360° VR luge video	Samsung Gear VR, Smartphone-based HMD	Visual, auditory	Open loop
Clifton and Palmisano (2020) [90]	Locomotion techniques in VR	Presence	Questionnaires, statistical analysis	IPQ, SSQ	None	Nature Treks VR	HTC Vive	Visual, auditory	Open loop
Servotte et al. (2020) [91]	Emergency training	Presence, stress	Questionnaires, statistical analysis	PQ, SSQ	None	Mass casualty incident	HTC Vive	Visual, auditory	Open loop
Saghafian et al. (2020) [92]	Safety training, fire emergency response	Presence	Questionnaires, statistical analysis	Custom questionnaire	None	Room-scale VR fire training scenario	HTC Vive Pro	Visual, auditory, passive haptic	Open loop
Škola et al. (2020) [93]	Virtual tourism	Engagement, presence	Questionnaire, statistical analysis	VR UX questionnaire	EEG	360° cultural heritage environments	HTC Vive	Visual, auditory	Open loop
Filter et al. (2020) [94]	Environmental education	Interest, joy, fear, presence	Questionnaires, statistical analysis	SPES, M-DAS	None	360° nature videos of wild wolves in their habitat	Oculus Quest, monitor	Visual, auditory	Open loop
Wang et al. (2021) [33]	Virtual reality locomotion	Presence	Questionnaires, statistical analysis	SSQ, IPQ	HR	Virtual mountain slope environment	HTC Vive	Visual, passive haptic	Open loop

Table 2. Cont.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Guo et al. (2021) [12]	Education	Positive and negative affect, presence	Questionnaires, statistical analysis	PQ, PANAS, SSQ	None	Educational VR content	HTC Vive Focus	Visual, auditory	Open loop
Grassini et al. (2021) [6]	VR-based procedural training and skill acquisition	Presence	Questionnaires, statistical analysis	PQ, SSQ	None	Warehouse training task (assembling model airplane) in VR Custom interactive VR news simulation based on SARS hospital investigative news Virtual counseling room with intelligent virtual agent (IVA)	HTC Vive Pro	Visual	Open loop
Wu et al. (2021) [95]	VR journalism	Empathy	Questionnaires, statistical analysis	TEQ, PANAS	None	Virtual toy truck assembly environment Interactive virtual chemistry laboratory Custom-designed VR music video environment	Oculus Rift	Visual, auditory	Open loop
Zuniga et al. (2021) [96]	Stress reduction	Presence, stress	Questionnaires, statistical analysis	PQ, SUS, Self-reported stress rating	HR	Virtual bar environment	HTC Vive	Visual, passive haptic	Open loop
Brade et al. (2021) [97]	VR training and simulation design	Presence, engagement	Questionnaires, statistical analysis	ITC-SOPI	None	Custom unity scene	HTC Vive	Visual	Open loop
Qorbani et al. (2021) [98]	VR safety training	Presence	Questionnaire	SUS	None	Simple indoor virtual room The Conjuring 2 VR teaser	Oculus Quest	Visual	Open loop
Pedersen and Nordahl (2021) [26]	Immersive media	Presence	Questionnaire, statistical comparison	SUS	Eye-tracking		HTC Vive Pro Eye	Visual, auditory	Open loop
Eiler et al. (2021) [9]	Addiction therapy	Presence, copresence, motivation, craving	Questionnaires, behavioral task performance metrics	IPQ, PQ, VRSQ	None		HTC Vive Pro	Visual, auditory, passive haptic	Open loop
Mayor et al. (2021) [99]	VR interaction/locomotion evaluation	Presence	Questionnaires, statistical analysis	IPQ, SSQ	None		HTC Vive	Visual, auditory	Open loop
Nakano et al. (2021) [100]	VR hardware design	Presence	Questionnaires, statistical analysis	IPQ	None		HTC Vive (modified)	Visual	Open loop
Song et al. (2021) [101]	Virtual reality exposure therapy	Anxiety, presence	Self-report, statistical analysis	TCI, STAI, ITC-SOPI	HR, EDA, skin temperature, RSP		HTC Vive	Visual, auditory	Open loop

Table 2. Cont.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Aseeri and Iterrante (2021) [102]	Communication and collaboration in virtual environments	Social presence	Questionnaire, statistical analysis	NMSPI	None	Collaborative VR interaction space	HTC Vive	Visual, auditory	Open loop
Magalhães et al. (2021) [103]	Cybersickness and presence research in VR navigation	Presence	Questionnaire, statistical analysis	IPQ, SSQ	None	Two virtual place experiences corresponding to real locations	Oculus Rift DK2	Visual, auditory	Open loop
Kim et al. (2021) [104]	VR quality of experience	Presence	Questionnaires, statistical prediction model	5-point single-item ratings	None	100 Unity-based VR videos	HMD not specified	Visual, auditory	Open loop
Morélot et al. (2021) [8]	Safety training	Presence	Questionnaire, statistical analysis	PQ, SUS	None	Interactive fire emergency scenario	Monitor	Visual, auditory, passive haptic	Open loop
Teixeira and Palmisano (2021) [105]	VR interaction design	Presence	Questionnaire, statistical analysis	SSQ	Postural metrics	Marvel Powers United VR	Oculus Rift CV1	Visual, auditory	Open loop
Gall et al. (2021) [106]	Affective VR, embodiment research	Embodiment, valence, arousal, dominance presence	Questionnaires, statistical analysis	SAM, Custom Embodiment Questionnaire	None	Minimalistic VR room with a virtual arm and table	HTC Vive	Visual, auditory, passive haptic	Open loop
Yung et al. (2021) [107]	Tourism marketing	Presence, valence, arousal	Questionnaires, statistical analysis	ITC-SOPI, SAM	None	Fully synthetic, interactive 3D model of a cruise ship	HTC Vive	Visual	Open loop
Brivio et al. (2021) [108]	Affective VR	Presence, anxiety	Questionnaires, statistical analysis	VAS, PANAS, SUS	HR	Relaxing nature environments	Smartphone-based HMD	Visual, auditory	Open loop
Shadiev et al. (2021) [109]	Cross-cultural education	Trait emotional Intelligence, presence	Questionnaires, qualitative analysis	TEIQue-SF, PQ	None	360° cross-cultural experience videos	Samsung Gear VR	Visual, auditory	Open loop
Gibbs et al. (2022) [110]	VR interaction research	Presence	Questionnaires, statistical analysis	IPQ	None	Virtual stick and ball-bouncing task	Oculus Rift S	Visual, active haptic	Open loop
Dunmoye et al. (2022) [111]	VR in engineering education	Social presence	Qualitative affective analysis using coded interaction indicators	None	None	Desktop VR land surveying simulator	Monitor	Visual	Open loop

Table 2. Cont.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Bayro et al. (2022) [112]	Remote collaboration in VR	Presence, copresence, arousal	Questionnaire	IPQ	EDA (NS-SCRs)	Spatial™ mixed reality platform	Oculus Quest 2, monitor	Visual, auditory	Open loop
Bayro et al. (2022) [113]	Presence research	Spatial presence	Questionnaire, statistical modeling	IPQ	None	Multiple virtual environments	Not specified	Not specified	Open loop
Muravevskaia and Gardner (2022) [114]	VR educational games	Social presence, empathy, fear, distress	Qualitative affective analysis	KEDS	None	VR Empathy Game	HMD not specified	Visual, auditory	Open loop
Wang et al. (2022) [115]	Collaborative VR training and teamwork analysis	Social presence, engagement, satisfaction and stress	Questionnaires	Custom collaboration and engagement questionnaire	ECG (HR, HRV)	Collaborative VR game	HTC Vive	Visual	Open loop
Wriessnegger et al. (2022) [116]	Presence research	Presence	Questionnaires, statistical comparison	IPQ, NASA-TLX	ECG (HR, HRV)	Four virtual rooms with different visual realism and lighting conditions	HTC Vive	Visual	Open loop
Tao et al. (2022) [117]	VR stress induction and affective computing	Stress, anxiety, presence	Supervised machine learning classification	SSAI	ECG (HR), EDA, Eye-blink rate	Three custom interactive VR stress scenes	HTC Vive Pro	Visual, auditory	Open loop
Ito et al. (2022) [31]	Multisensory VR	Presence	Questionnaire, statistical analysis	IPQ	None	Custom VR underwater scene	HTC Vive Cosmos	Visual, auditory, thermal haptic	Open loop
Jun et al. (2022) [118]	VR media psychology	Arousal, presence	Questionnaires, behavioral metrics, statistical analysis	SAM, Custom 3-item presence scale	Head movement tracking	360° real-world videos	HTC Vive	Visual, auditory	Open loop
Meirinhos et al. (2022) [119]	VR marketing	Presence	Questionnaire, statistical analysis	IPQ	None	Virtual product showroom	HTC Vive Pro	Visual	Open loop
Hernández et al. (2022) [120]	Presence control in VR systems	Presence	Real-time computational estimation, control theory	None	ECG (HR)	Park, height exposure environment	Oculus Rift	Visual, auditory, active haptic	Closed loop
Miguel and Hartmann (2022) [121]	Social VR	Spatial presence, social presence	Questionnaires, statistical analysis	NMSPI, SPES	None	Social VR platforms	Not specified	Not applicable	Not applicable

Table 2. Cont.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Melo et al. (2022) [30]	Virtual tourism	Enjoyment, presence	Questionnaires, statistical analysis.	IPQ, Enjoyment scale, Positive affect scale	None	VR reconstructed São Leonardo da Galafura viewpoint (Portugal)	HTC Vive	Visual, auditory, passive haptic, olfactory	Open loop
Lemmens et al. (2022) [122]	VR gaming, affective VR	Fear, hostility, enjoyment, presence, arousal	Questionnaires, statistical analysis	MEC-SPQ, Fear scale, Enjoyment scale, State Hostility Scale [123]	HR, HRV LF/HF	Resident Evil 7 VR, Doom VR	PlayStation VR, monitor	Visual, auditory	Open loop
Mizuho et al. (2023) [124]	VR cognitive research	Presence	Questionnaire, statistical analysis	IPQ	None	Virtual replica of a real room	Meta Quest 2	Visual	Open loop
Westermeyer et al. (2023) [125]	Interaction research	Presence (spatial and plausibility dimensions)	Questionnaires, statistical analysis	IPQ, Custom Plausibility Questionnaire	None	Interactive room	Varjo XR-3	Visual	Open loop
Covaci et al. (2023) [126]	Cinematic VR	Presence	Questionnaire, statistical analysis	SUS	None	360° videos	Smartphone-based HMD	Visual, auditory, olfactory	Open loop
Mal et al. (2023) [127]	Embodiment research in VR	Spatial presence, embodiment	Questionnaires, statistical analysis	VHPQ, VEQ, IPQ, SSQ	None	Sports fitness room, office environment	Valve Index HMD	Visual	Open loop
Llinares et al. (2023) [128]	Environmental psychology	Presence	Questionnaire, statistical analysis	SUS	EEG, HRV (HF), EDA	Replicated classroom environment	HTC Vive	Visual, auditory	Open loop
Salminen et al. (2023) [129]	Neurofeedback-based VR for meditation training	Presence	Questionnaires, statistical analysis	ITC-SOPI, MEDEQ	EEG	“RelaWorld”	Oculus Rift DK2	Visual, auditory	Closed and Open loop
Li and Kim (2024) [130]	Virtual workspace design	Presence, positive and negative affect	Questionnaires, statistical analysis	IPQ, PANAS	ECG (HR)	Simulated virtual home office workspace	HMD not specified	Visual, auditory, olfactory	Open loop
Chittaro et al. (2024) [131]	Anxiety and stress reduction	Presence, relaxation, stress	Questionnaires, statistical analysis	IPQ, STAI, PANAS	HR, EDA, RSP	Natural coastal VR environment	Meta Quest 2	Visual, auditory	Closed loop
Maymon et al. (2024) [132]	Emotion induction in VR	Fear, presence	Questionnaires, statistical analysis	PQ, IPQ, DEQ, AQ, ITQ, ERQ, SSQ	ECG (HR), EDA (SCL)	Simulación VR “Richie’s Plank Experience”	HTC Vive	Visual, auditory	Open loop
Coelho et al. (2024) [133]	Virtual training	Presence	Questionnaires, statistical analysis	IPQ, SSQ	None	Virtual workshop	HTC Vive Pro	Visual	Open loop

Table 2. Cont.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Oliveira et al. (2024) [134]	Firefighter training	Positive and negative affect, presence, engagement	Questionnaires, statistical analysis	PANAS, ITC-SOPI, DASS-21	ECG (HR), EDA (SCL), EEG	FLAIM Trainer VR	HTC Vive Pro	Visual, auditory, passive and active haptic	Open loop
Safikhani et al. (2024) [135]	Presence research	Presence	Questionnaire, statistical analysis	IPQ	EEG	Two desert scenes	Meta Quest 2	Visual	Open loop
Chen et al. (2024) [136]	Cognitive rehabilitation	Presence	Questionnaires, statistical analysis	PQ	HR, HRV, EDA	Unity-based everyday scenarios	HTC Vive Cosmos	Visual	Open loop
Pears et al. (2024) [137]	Healthcare education	Presence	Questionnaires, statistical analysis	SUS	None	Three VR reusable e-resources Virtual training replica of real firefighting door-opening procedure	Smartphone-based HMD	Visual, auditory	Open loop
Narciso et al. (2024) [138]	Firefighter training	Stress, fatigue, presence	Questionnaires, statistical analysis	IPQ, SSQ, PSS, CIS, VAS	ECG (HRV LF/HF)	Boardroom and Auditorium scenarios	HTC Vive Pro	Visual, auditory, thermal haptic	Open loop
Spyridonis et al. (2024) [139]	VR exposure therapy for public speaking anxiety	Anxiety, presence	Questionnaires, statistical analysis	PRPSA, IPQ	None	VROOM (Virtual Reality for the Observation of Oncology Models)	Oculus Rift S	Visual, auditory	Open loop
Gronowski et al. (2024) [140]	Data visualization usability	Presence	Questionnaire, statistical analysis	IPQ	None		Oculus Quest 2	Visual	Open loop
Pavic et al. (2024) [141]	Emotion induction in VR	Valence, arousal, presence	Questionnaires, statistical analysis	SAM, PANAS, SPES, TPI-SR, HADS	HR, EDA (SCL)	360° videos	Samsung HMD Odyssey, monitor	Visual, auditory	Open loop
Berni et al. (2024) [142]	Product and architectural design evaluation in VR	Presence	Questionnaire, statistical analysis	Single-item Presence rating	Eye-tracking	360° images of a real tiny-house interior	HTC Vive	Visual	Open loop
Shah and Lawson (2025) [143]	Height-exposure VR	Fear, presence	Questionnaires, statistical analysis	IPQ, Custom emotion questionnaire	HR	“Richie’s Plank Experience”	Pico 4 Enterprise	Visual, auditory, passive haptic	Open loop
Bayro et al. (2025) [144]	Remote collaboration in VR	Arousal, presence	Questionnaire, statistical analysis	IPQ	ECG (HRV, RMSSD), EDA (NS-SCRs)	Spatial.io shared 3D workspace	Meta Quest 2, monitor	Visual, auditory	Open loop
Ronca et al. (2025) [145]	Presence assessment in VR	Presence, arousal	Questionnaire, statistical analysis	Custom sense of presence questionnaire	EEG; ECG (HR, HRV), EDA (SCL)	Racetrack driving environment	Oculus Rift DK2	Visual, auditory, passive haptic	Open loop

Table 2. Cont.

Research	Application Field	Affective State	Affective Recognition Approach	Questionnaires	Physiological Signals	Virtual Environment	Display	Stimuli Type	Control Approach
Ribé et al. (2025) [146]	VR exposure therapy for acrophobia	Anxiety, fear, presence	Questionnaires, statistical analysis	AQ, SUS, STAI	None	“Top Floor” height scenario	HTC Vive	Visual, auditory, passive and active haptic	Open loop
Khundrakpam et al. (2025) [147]	Stress elicitation and assessment in VR	Stress, presence	Questionnaires, statistical analysis	PSS, SSQ	HR, HRV, EMG	bWell platform	HTC Vive Pro Eye	Visual, auditory	Closed and Open loop
Archer et al. (2025) [148]	Emotion/stress manipulation in narrative VR	Arousal, stress, presence	Questionnaires, statistical analysis	SSQ, Custom questionnaire	HR, HRV LF/HF, EDA, skin temperature	London bus narrative environment	Meta Quest 2	Visual, auditory	Open loop
Naud et al. (2025) [149]	Disaster risk and evacuation research	Stress, presence	Questionnaire, statistical analysis	IPQ	HR; EDA (ISCR)	Tsunami scenario	HTC Vive Pro	Visual, auditory	Open loop
Pannattee et al. (2025) [150]	Presence assessment	Presence	Questionnaire, statistical analysis, machine learning	IPQ	None	Eight Unity scenes	Oculus Quest Pro	Visual	Open loop

4.1. Trending Presence Factors

Section 2 notes that factors influencing the generation of presence can be divided into external and internal categories. Several investigations have examined specific factors influencing presence. Some of these works drew conclusions about these factors from their experimental results, but not with the initial intention of assessing them.

Given the relationship between presence and immersion, some researchers have investigated factors that contribute to immersion in virtual environments, aiming to enhance presence. The objective is to capture the user's attention while ensuring an enjoyable experience. As mentioned earlier, the primary expected outcome is that immersive environments can increase presence. This has been validated on the results of [25,26,77,83,92,103,126,151]. In summary, these results demonstrate that detailed geometries, richer textures, and enhanced sound quality can effectively evoke presence.

Later, the results of [111,112,121] focused on social interaction, aiming to elicit a sense of presence through coexistence with other virtual avatars. Moreover, these articles aim to generate realistic, immersive environments that enable remote collaboration. These results were largely motivated by the COVID-19 pandemic to overcome confinement [152], to observe benefits for mental health [121], and to implement techniques such as mindfulness for stress reduction [153]. Additionally, Ref. [79] argues that presence was enhanced when the virtual reality experience evoked authentic cognitive, emotional, and behavioral responses, and when participants were able to construct their own narratives of the events.

Another factor influencing presence found in the literature is the subject's individual characteristics. Some studies attempt to draw general conclusions on subject characteristics like sex [81,113,119], age [28], or personality features [89,91,101]. Since this area has been less thoroughly explored, the conclusions are not yet broadly established. Research indicates that men and women experience presence at comparable levels, with the primary differences related to age, as older individuals tend to report higher levels of presence [28]. In terms of experience, men often report higher levels of presence, not necessarily due to the genre, but because they tend to have prior exposure to VR or gaming environments [81].

Furthermore, a few studies have investigated perceived control and freedom of movement in subjects immersed in a virtual reality environment [7,51,76,90,154]. Their findings suggest a positive correlation with presence, highlighting the importance of improving these immersive elements. Another studied factor associated with the virtual environment is the field of view (FOV) displayed to users. It has been explored how changing the FOV affects presence. The predominant evidence suggests that expanding the FOV up to a certain threshold enhances presence [100,154]. Nonetheless, excessive expansion may induce symptoms of cybersickness, which reduce the sense of presence. An interesting approach to mitigating this issue is the dynamic restriction of the visual field [105].

One of the presence illusions, embodiment, has been studied in the literature to evoke presence. Studies on embodiment not only analyze the impact of the virtual body on presence but also the manner of representation that best supports its purpose [73,102,106,127]. For instance, Ref. [102] observed that the highest levels of co-presence are achieved when a completely virtual avatar is used, unlike avatars created from scanners. The same article suggests this may be due to the greater availability of facial expressions and other subtle behavioral cues, which can convey more emotion and more comprehensive thoughts. Another important example is the results presented in [127], which studied congruence in the representation of the virtual reality avatar. By manipulating variables such as gender and clothing, they concluded that the avatar–environment congruence significantly affected the avatar's plausibility but not the sense of embodiment or spatial presence.

It is important to note that not all the assessed studies in this review focus on a single aspect of presence, or on all aspects; only a subset has been cited above. This illustrates how only a small set of authors dedicated their research to this particular area.

4.2. Stimulus Signal Type

One key aspect of achieving presence in VR is the effectiveness of the stimuli in the virtual environment. Currently, researchers are studying the relationship between different types of virtual stimuli and presence and how these stimuli can enhance user experiences. Virtual reality stimuli can include visual, auditory, haptic (tactile and kinesthetic), and, in rare cases, olfactory signals. It is essential to understand how the quality, quantity, and timing of these stimuli influence the user's perception, and how interactions among different types of stimuli can further enhance presence. However, these details are rarely reported in the literature, providing only general information about the type of environment used and which signals are included.

The ninth column of Table 2 includes the data related to the types of stimulus signal used to provide sensory feedback to users. The main tendency is to combine visual and auditory stimuli, which is reasonable since this can be easily achieved with a single head-mounted display. The second most common approach uses only visual stimuli, while the third combines visual, auditory, and haptic stimuli, which has been demonstrated to generate higher levels of presence due to the impact of haptic stimuli [31,32,87,91,106,120,155]. It is essential to note that implementing haptic stimuli requires specialized hardware, software, and control elements to ensure proper signal synchronization. If this synchronization fails, and haptic signals are not rendered on time (kinesthetic signals up to 100 Hz, vibrotactile signals up to 1000 Hz), the system may have negative effects on users, such as motion sickness, inaccurate interactions, and interrupted presence [156].

In addition, some works report the use of visual signals without specifying whether audio was included [83,97,98,102,115,116]. Despite their findings, the lack of clarity in describing the experimental setup makes it difficult to reproduce the same experimental scenario to test and explore them. Only a few studies have explored the integration of olfactory signals [27–30], using burning-scent, food, and beverage scents. A remarkable study presents an olfactory device integrated with HTC Vive controllers [29], demonstrating the feasibility and potential of these underexplored signals beyond therapeutic applications.

Another important element to note in the reviewed literature is the discussion by some authors regarding temperature-based stimuli. These signals belong to haptic feedback [156], whose implementation is not straightforward (as visual or auditory). For instance, Ref. [31] uses temperature stimuli (specifically cold) in a virtual underwater environment, demonstrating positive effects on achieving higher levels of presence. Another research that studies temperature as a stimulus was presented by [32], who developed a haptic vest that allows temperature to be manipulated in a simulation of a railway disaster. Moreover, Ref. [100] highlights the negative impact of the absence of heat stimuli on their test subjects. In their fire simulation, the lack of heat feedback led subjects to report a mismatch between expected and perceived stimuli. This example remarks the importance of new feedback signals, while also highlighting the challenges of reproducing diverse stimuli in VR systems.

To summarize the frequency of use of each stimuli combination in the review studies, Figure 3 depicts a pie chart representing the percentages of studies using each type of stimulus combination. This percentage is based on 76 articles (those with experimental results), of which 77% involve visual and auditory stimuli (or its combination). Another 15% corresponds to the most praised combination, visual + auditory + haptic, to evoke presence. Meanwhile, other combinations, including olfactory cues, account for 8% or less.

This disproportionality may be associated with the technological complexity required to generate these feedback signals. However, no discussion of how each study selects its stimulus cue type is presented.

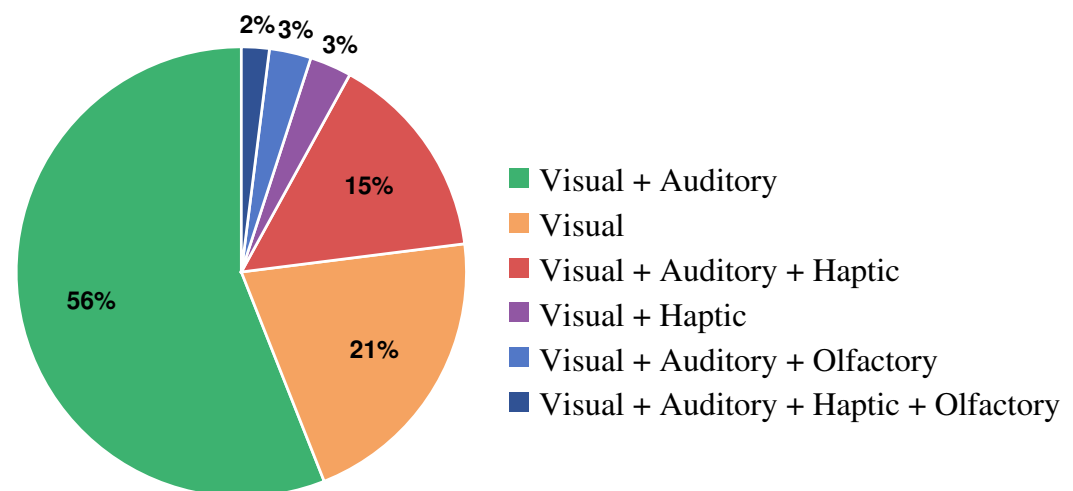


Figure 3. Distribution of sensory stimulus combinations in VR presence assessment studies.

4.3. Main Display Devices

Nowadays, a wide variety of technological devices are used in VR implementation. The research community focuses on evaluating their ability to create vivid experiences that evoke presence. In general, a detailed study of the devices themselves (e.g., screen quality) is not conducted, since their manufacturers handle this. Instead, the focus is on understanding how these devices affect users' perception of virtual environments and comparing them to determine which ones evoke presence more effectively.

Primarily, the use of head mounted displays (HMDs) is compared to computer monitors, with the former demonstrating superior performance to evoke presence [69,70,78,82,112]. This is a logical result, since HMDs occlude real surroundings, allowing users to concentrate on the device's FOV. Moreover, the stereoscopy feature of HMDs provides depth sense, which is impossible to achieve with single screens. Therefore, the natural path to follow is to enhance or optimize HMDs to improve user experiences. For instance, Ref. [100] modified an HMD (specifically an HTC vive) by extending its FOV. Their results suggest that increasing the field of view improves the presence and self-location. However, some cautions must be taken to avoid cybersickness.

Over the past few years, the HTC VIVE family has been the most widely used HMD, followed by the Oculus Rift. Some researchers have implemented newer variants like the Oculus Quest or Meta Quest; however, HTC remains the most popular, offering a broad range of devices, although at a higher cost. Other devices, such as Sony HMD, Valve Index, or Playstation VR, are present on the video game market, but without being used for research due to more requirements during rapid prototyping of the VR systems. Conversely, some works still use the classic approach of computer screens to evaluate specific features of their virtual environments. The least popular display device at present is the CAVE system, which requires a considerably higher monetary investment and involves deploying specialized hardware.

The eighth column of Table 2 includes the displayed devices used in each analyzed manuscript, while Figure 4 illustrates their use as percentages. From the pie chart, 85% of the experimental setups included a head mounted display, while 13% still use monitor setups. This last is questionable since monitors' immersive capabilities are very limited. Finally, only 2% of the studies still implement fully immersive cave systems.

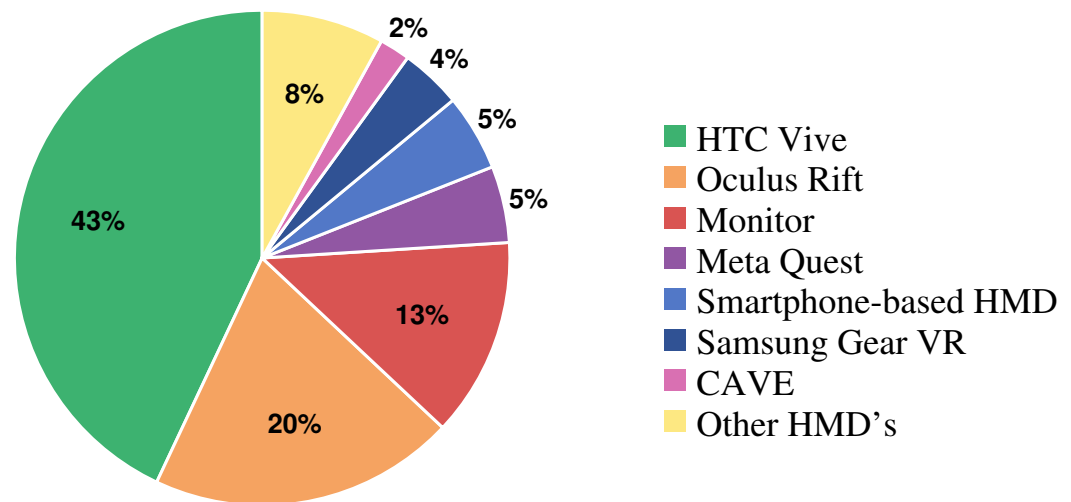


Figure 4. Display devices employed in VR presence assessment studies.

4.4. Presence Assessment

Assessing presence has become fundamental in evaluating VR systems, providing valuable insights into user experience quality and engagement with the virtual environment. This has guided the development and continuous improvement of VR systems. However, it remains challenging, as presence corresponds to a psychological state. Therefore, research on assessing and quantifying presence remains an open problem, and a large number of multidisciplinary research groups have continuously debated and proposed methods to address it. Overall, the literature identifies two general approaches to assessing presence: subjective and objective.

The subjective approach considers metrics collected from users through surveys, interviews, or self-reported data on Likert scales, in which participants evaluate their VR experience, typically after the session. Conversely, the objective approach focuses on the acquiring and processing of physiological responses, which are then interpreted to estimate psychological responses and affective states. Unlike subjective measures, physiological responses are recorded online during the VR experience, enabling more precise analysis of user responses as stimuli are presented.

Below, a more precise description of the current subjective and objective instruments to assess presence will be presented. It is important to note that there are questionnaires designed to assess cybersickness, which does not evaluate presence. However, it provides valuable insights into the level of discomfort experienced by users, which helps researchers to determine the extent to which such discomfort may have disrupted presence.

4.4.1. Subjective Assessment

Among the subjective instruments to assess presence in virtual reality studies, the dominance of four presence questionnaires and one cybersickness evaluation instrument is remarkable and is consistent with previous reports in the literature (e.g., [39,41]). Other presence questionnaires exist, introduced and tested on specific articles, but no continuity in their use has been observed. Additionally, there exist affective states questionnaires and others to assess immersion, virtual body ownership, embodiment, or plausibility. Still, their use is minimal, making determining their effectiveness difficult. The fifth column of Table 2 includes the questionnaires implemented in each study (where applicable). The choice of which questionnaire depends mainly on the extent of information the researcher aims to extract regarding a specific feature of the VR experience, which varies according to the diverse set of questions in each questionnaire. In addition, selection is influenced by the

specific conclusions the researcher intends to draw from the research and its subsequent applications. Below, the questionnaires most frequently used in Table 2 are described.

1. **Slater–Usoh–Steed (SUS) questionnaire:** This questionnaire, presented by Slater, Usoh, and Steed in 1994 [157], consists of six questions addressing three aspects of physical presence in a virtual environment (VE). The three items include the user's sense of being in the VE, the extent to which the VE becomes the dominant reality, and the extent to which the virtual environment is remembered as an actual place.
2. **Presence Questionnaire (PQ):** In 1998, Witmer and Singer identified involvement and immersion as the two necessary conditions for presence [158]. Consequently, they proposed a questionnaire to assess the presence of these two conditions by identifying the factors influencing them. Initially, the questionnaire comprised 32 items, and the factors were categorized into three categories: involved/control (perceived control over the events of the VE), natural (the degree to which interactions feel natural and the level of consistency between the VE and reality), and interface quality (how control and display devices either disrupt or distract users and the degree of concentration that users can devote to the tasks in the VE). In 2005, the authors modified the questionnaire [159], reducing the number of items to 29 and dividing the factors into four categories: involvement (the degree to which the user focuses their energy and attention on a coherent set of stimuli or meaningfully related activities and events), adaptation/immersion (the perceived proficiency of interacting with and operating in the VE and how quickly the user adjusted to the VE experience.), visual fidelity (the degree to which the VE configuration permits active search or examination of the objects in the VE using vision), and interface quality (the perceived quality of the VE interface and the extent to which it does not interfere with activities in the VE).
3. **Igroup Presence Questionnaire (IPQ):** Schubert, Friedmann, and Regenbrecht [160] postulated that presence develops from constructing a spatial-functional mental model of the virtual environment, involving two cognitive processes: construction, or the representation of bodily actions as possible actions in the VE, and suppression of incompatible sensory input. They proposed that the conscious presence encompasses two key components: spatial presence and involvement. To investigate these components, they integrated items from previously published questionnaires (Witmer and Singer [158], Hendrix [161], Carlin, Hoffman and Weghorst [162], and Slater–Usoh–Steed [157]) with newly proposed items into a unified questionnaire. The latest version of this questionnaire comprises 14 items. The IPQ is organized into three subscales and one additional general item not belonging to a specific subscale: spatial presence (the sense of being physically present in the VE), involvement (measuring the attention devoted to the VE) and experienced realism (measuring the subjective experience of realism in the VE). The extra item, the general presence, determines the overall subjective sense of being in the VE by averaging the mean scores of the spatial presence, involvement, and experienced realism subscales.
4. **ITC-presence Inventory (ITC-SOPI):** In 2001, Lessiter et al. introduced a questionnaire focusing on users' subjective experiences of media, without reference to objective system parameters [163]. Initially, the questionnaire comprised 63 items divided into two parts, A (7 items) and B (56 items), addressing participants' experiences before and during the mediated environment, respectively. The primary objective was to encompass attributes considered relevant for defining presence based on the existing literature, including sense of space, involvement, attention, distraction, control and manipulation (autonomy), realness, naturalness, perception of time, awareness of behavioral responses, a sense of social interaction (parasocial and copresence), personal relevance, arousal, and adverse effects. Following a series of analyses, the authors

presented a revised and reduced version of the questionnaire, totaling 44 items distributed across two parts: part A consists of 6 items, and part B comprises 38 items. Four factors were identified: sense of physical space (the feeling of being physically in the simulated environment, being able to physically control and manipulate elements of the VE, 19 items), engagement (the psychological involvement, interest, and user satisfaction with the content of the experience, 13 items), ecological validity (realism of the content and the naturalness and consistency of the VE, 5 items), and negative effects (the adverse psychological reactions of the immersion).

5. **Simulator Sickness Questionnaire (SSQ):** Although not exclusively designed as a presence questionnaire, it is frequently employed with presence measures to evaluate the incidence of simulator sickness, a factor that can impact the overall presence. Developed by Kennedy et al. [164], the SSQ quantifies the severity of symptoms associated with simulator sickness. The questionnaire covers three main symptom clusters:
 - Oculomotor symptoms—these include symptoms such as eyestrain, difficulty focusing, and blurred vision.
 - Disorientation symptoms—these encompass feelings of dizziness, vertigo, and spatial disorientation.
 - Nausea symptoms—symptoms related to nausea and general discomfort, such as headache and increased salivation.

Participants must complete the SSQ before, during, and after exposure to a simulator or virtual environment. Subsequently, the scores are analyzed to determine the level of discomfort or sickness experienced by the participants. Researchers and developers of virtual environments utilize the SSQ to evaluate the impact of virtual reality setups, applications, or experiences on users' well-being. The SSQ comprises 16 items.

To better understand the frequency of presence questionnaire usage, Figure 5 shows the number of studies in the reviewed literature that use each presence questionnaire. The preference for the IPQ [160] is clear, while PQ, SUS, and ITC-SOPI have been used three or more times (in recent years) and the rest of the questionnaires have been used one or two times maximum (sometimes by the same authors). Cybersickness and other types of questionnaires are not listed in Figure 5 since their objectives are not directly to assess presence but to serve as a complementary tool along with other approaches.

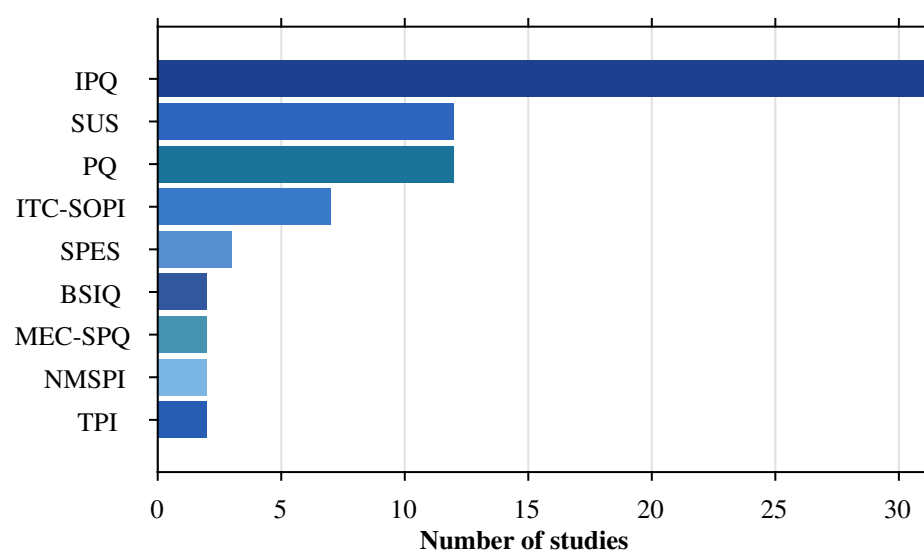


Figure 5. Incidence of the use of questionnaires for subjective presence assessment.

Further, it is worth noting that affective questionnaires are rarely used alone (unlike presence questionnaires), as they primarily aim to correlate responses with affective states

estimated from physiological changes. Among questionnaires of this kind, the most popular and widely studied is the Self-Assessment Manikin (SAM) [165]. The SAM evaluates user's responses to a stimulus in terms of arousal, valence, and dominance using a non-verbal pictorial assessment tool. The SAM is an instrument widely used in affective computing to evaluate a wide range of stimuli and multimedia elements, and most of the results fall within this computer science discipline (e.g., [166,167]).

4.4.2. Objective Assessment

Objective measures of presence assessment have gained attention and popularity among researchers as complements to traditional questionnaire-based evaluations. The sixth column of Table 2 summarizes studies that use physiological signals to objectively assess presence, with cardiac and electrodermal activity as the most popular. Despite the significant information that EEG can provide for evaluating user perception of virtual environments, its analysis is performed exclusively in the frequency domain, making its implementation on embedded systems or for real-time analysis difficult. To complement the table data, Figure 6 exhibits the number of studies that use each physiological signal alone or in combination with other signals. Again, the most widely used signal is ECG, which combines EDA, EEG, and RR.

Physiological signals are processed to obtain time- and frequency-domain characteristics, which are used in diverse affect-detection methods [60,166]. Several approaches exist for obtaining and evaluating objective metrics, depending on the application, technology, and devices used in the VR system. One of the most widely used approaches is to identify statistically significant differences (hypothesis testing) in physiological signal characteristics that indicate the evocation of different affective states. Various statistical analysis techniques are employed for this purpose, with the choice of technique largely determined by the data collected. If normal data are obtained, it is common to follow up with ANOVA tests; otherwise, non-parametric techniques have to be considered for such a purpose (e.g., Mann–Whitney U test, Friedman Test, Wilcoxon signed-rank test, Spearman rank-order correlation).

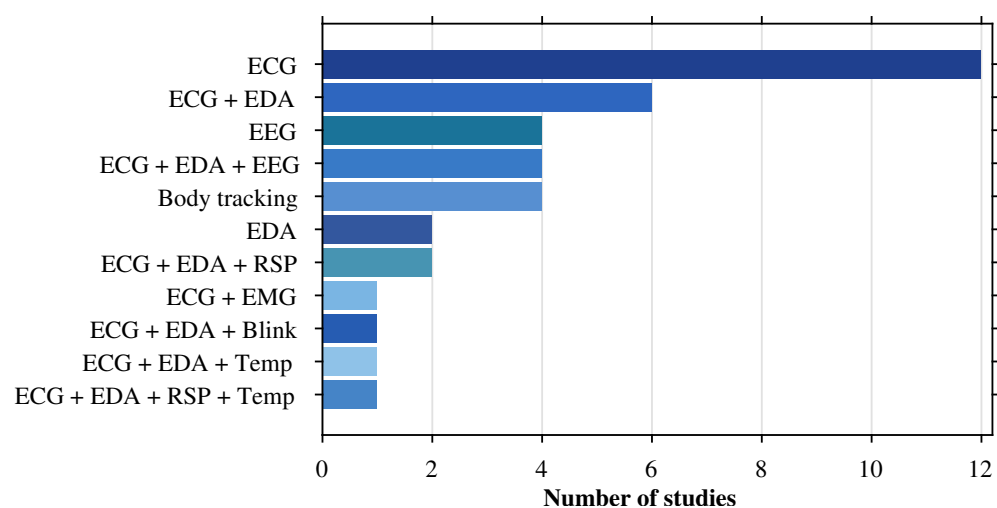


Figure 6. Number of studies using single or combined physiological signals for affect detection.

Furthermore, in affect analysis, two types of approaches can be distinguished. The first corresponds to discrete emotions (DEMs), which are innate, universal, and present in every person regardless of ethnicity or cultural differences [168]. The literature mentions joy, sadness, anger, fear, disgust, and surprise among the most popular DEMs. In contrast, the second approach treats affective states as continuous. According to this viewpoint, emotions are not limited to a finite and specific set of categories but are fluid and can vary

in intensity, duration, and combinations. The continuum-of-emotions approach suggests that affective states can be better understood as emotional states that vary in degree and intersect itself. For example, rather than simply feeling “happy” or “sad,” a person might experience a range of emotions from extreme joy to mild sadness, depending on the context and circumstances. Researchers studying affective states under this approach employ the theory proposed by psychologist James Russell and their Circumflex model of emotion [169]. This model suggests that emotions can be represented in a two-dimensional space, with two main dimensions:

- **Valence:** This dimension refers to the degree of pleasure or displeasure associated with an emotion. Positive emotions, such as happiness and joy, are at the positive end of this dimension, while negative emotions, such as sadness and anger, are at the negative end.
- **Arousal:** This dimension refers to the physiological activation or energy level associated with an emotion. High-arousal emotions, such as anger, fear, or excitement, are at the high end of this dimension, while low-arousal emotions, such as calmness and boredom, are at the low end.

Additionally, a third dimension, called dominance, was incorporated to the model to describe the degree of control experienced during an emotion. The dominance dimension distinguishes between emotions that imply a sense of control or power (e.g., pride, confidence) and emotions that imply a sense of submission or lack of control (e.g., shame, anxiety). Studies employing only physiological signals for affect detection commonly do not include the dominance dimension, whereas studies incorporating other tools, such as the Self-Assessment Manikin (SAM), take into consideration dominance to obtain more detailed results.

While both the discrete and continuous emotion approaches offer different perspectives on the nature of affective states, it is essential to recognize that both methods can provide valuable insights and complement each other in understanding human emotional experience. As mentioned before, studying affective states has been the most objective alternative for analyzing presence. Nonetheless, there is still a long way to go to understand with greater certainty how certain stimuli signals and virtual environments (scenarios and activities) relate to specific affective states.

The third column of Table 2 includes the affective states estimated on each study, while the seventh column presents the virtual environments and scenarios implemented on each experimental setup. Both elements are strongly related, since the estimated affective states depend on the type of environment used. It will be impossible to estimate a specific affective state if the target of the virtual environment used is not known, due to the non-bijection property of the physiological responses. For example, similar high arousal levels could appear on scary environments as well as in an intense video game.

It has been observed that negative emotions seem to generate higher levels of presence or generate it more efficiently. For instance, studies focused on fear [25,80,95], stress [32,85,87,91,96,115,117] and anxiety [69,79,82,101] report clear conclusions designing phobia treatment or scary scenarios. One of the most significant contributions to the study of affective states concerning presence is the work by Riches [79], who conducted a qualitative analysis to identify the factors influencing presence. By examining these factors, Riches et al. established connections between specific emotions and presence, demonstrating that paranoia, loneliness, and self-recognition enhance presence. In contrast, emotions such as detachment tend to diminish it.

Further, some studies draw conclusions about the dimensions of emotions. For instance, [82] states that the highest arousal levels are obtained when participants are exposed to negative stimuli, suggesting that achieving high arousal with positive emotions may be

more difficult. This can explain the reason why researchers tend to work with negative emotions such as anxiety, stress, and fear. Evidence shows that scenarios that work with these emotions, such as disaster events, efficiently generate presence [78,91]. Conversely, negative stimuli result in low levels of valence, which is congruent with its definition. The importance of affective analysis in virtual reality systems is that the highest levels of presence are related to the most intense emotional states, where the arousal indicator becomes more relevant. Something interesting to mention is that even a low-immersive system may evoke high levels of presence if the VR scenario can engage high arousal emotions [80]. As for dominance, although it is the newest and least explored dimension, studies have found that the presence of a virtual body and its customization tends to increase it [73].

Integrating affective analysis in assessing presence offers a more comprehensive understanding of user experience in virtual reality, while presence focuses on the perceptual and cognitive aspects of the virtual reality experience, affective analysis captures the emotional responses that shape and reinforce this presence. By leveraging physiological and behavioral indicators, researchers can evaluate VR system effectiveness and enhance them by dynamically adapting content to the user's emotional state. This synergy has significant implications for fields where emotionally engaging experiences are crucial.

These indicators enable a more precise and continuous assessment of presence, allowing researchers to better understand user interactions with virtual environments. By integrating objective and subjective measures, a more comprehensive and reliable presence assessment can be achieved, leading to improved virtual reality experiences and more rigorous scientific findings. Since subjective metrics have been previously discussed, the rest of this section will focus on objective methods.

4.5. Cybersickness

Alongside VR's potential benefits for creating a variety of indoor experiences, there is a crucial challenge affecting users' presence: cybersickness. This condition encompasses symptoms such as disorientation, fatigue, dizziness, headache, increased salivation, dry mouth, eye strain, vomiting, stomach awareness, pallor, sweating, and postural instability, all of which are caused by the use of HMDs and stimulus mismatch when using VR [170]. Cybersickness diminishes the overall quality of the VR experience and even deters users from prolonged use.

While a strong sense of presence can enhance engagement, it may also exacerbate disorientation and discomfort, increasing susceptibility to cybersickness. Therefore, the evidence suggests that presence and cybersickness are negatively related. Understanding how these variables interact is crucial for researchers developing VR technologies that maximize presence while minimizing adverse effects on users' well-being. Moreover, Weech et al. provide a comprehensive review of these variables and summarize their observations in four points [154]:

1. Strategies that minimize sensory mismatch demonstrate potential for reducing cybersickness and enhancing presence.
2. Both presence and cybersickness are augmented by stereoscopy effect, high field-of-view display conditions, and by increasing the likelihood of inducing vection.
3. Enhancing factors such as interaction and the control of navigation results in higher presence and lower cybersickness.
4. Men and individuals with greater gaming experience exhibit lower cybersickness and higher presence, although the specific effects of gender and gaming experience are unclear.

Other researchers have also investigated the variables that correlate presence with cybersickness and explored alternatives to mitigate its adverse effects. For instance, Ref. [154] reports a negative correlation between the presence of a rich narrative and cybersickness. This is because incongruent sources of self-motion information, that is, the mismatch between what the user observes and the cues detected by the vestibular system, stimulate a specific pathway that involves the emetic component of vestibular nuclei [154]. These results are congruent with those obtained by Riches et al. [79]. Conversely, Mayor et al. [90,99] studied how interaction and locomotion can be worked in virtual reality systems. Their findings suggest that the Room Scale paradigm is most effective at evoking presence and reducing cybersickness.

The impact of the visual field has also been extensively studied to elucidate its effects on presence and cybersickness. Literature results exhibit that expanding the visual field enhances the presence, but it can also lead to the onset of cybersickness symptoms in users. To address this duality, Teixeira and Palmisano [105] proposed a potential solution by investigating the effects of dynamic visual field restriction, demonstrating that this technique effectively reduces cybersickness.

As measuring presence is essential for virtual reality systems, assessing cybersickness is also relevant. Venkatakrishnan et al. [86] provide a summary of the literature on cybersickness and detail that physiological measures have been reported as valid indicators for cybersickness: particularly, researchers have suggested that cybersickness is related to an increase in heart rate. Similarly, it has been argued that skin electrical conductivity varies during cybersickness, so skin conductance levels (SCLs)/electrodermal activity (EDA) also serve as valid indicators. They further note that other measurements, such as heart rate, blood pressure, volume, and skin temperature, are also associated with cybersickness. Given the need to evaluate cybersickness, proposals have been created, including Kim et al. [104], who studied the relationship between presence and VR sickness, developing a model capable of predicting these symptoms with an accuracy of 90%. The authors comment that their model can be used to generate content in virtual reality, aiming to reduce cybersickness recurrence and create more comfortable experiences.

5. Discussion

The previous literature synthesis reveals specific elements that influence presence. Although none are clearly predominant across the results, some aspects are highlighted by their positive results, particularly those associated with increased user presence, such as immersion, embodiment, and FOV. Immersion plays a key role in plausibility by providing users with vivid experiences. Results evidence the impact of detailed textures, geometries, and the overall quality of virtual environments on presence. However, overall conclusions indicate uncertainty about the true effect of high-fidelity environments, as the system's interactive level can also contribute to increased presence. This is consistent with subjective evaluations of virtual environments in the literature outside presence evaluation [36].

Conversely, factors such as interaction, field of view, and embodiment can increase presence when VR tasks have well-defined objectives, avatars are used (to evoke body ownership), and HMDs avoid excessive FOVs to prevent cybersickness. These factors are primarily determined by software and task planning, and overall conclusions indicate that simply integrating more and higher-technology elements is not the best way to achieve greater user presence. This is a significant concern, as incorporating more technology and feedback elements could substantially increase costs and the complexity of controlling and synchronizing the complete VR system. Therefore, researchers must focus on providing plausible, interactive, and well-structured virtual environments that include an avatar (preferably with body tracking) to achieve higher levels of presence.

Furthermore, the revision of the stimuli signals type reveals that 77% of experimental research uses visual and auditory signals, whereas few studies include haptic feedback and even fewer include olfactory feedback that has led to a conflicting discussion. First, the use of visual and auditory signals follows the fact that more feedback elements are not strictly necessary to generate presence. This is consistent with the widespread use of HMDs, which enable the integration of visual and auditory signals within a single system. However, the lack of haptic or olfactory signals may indicate that the full potential of VR remains unexplored, as more realistic experiences have yet to be achieved. Some evidence, including visual, auditory, and haptic feedback, has shown that higher levels of presence can be generated, making its implementation desirable. Researchers may begin integrating new feedback elements into their VR systems, targeting simple scenarios that do not require complex, highly costly feedback systems. For instance, low-degree-of-freedom motion systems can be developed [120], rather than acquiring expensive treadmills or 6-degree-of-freedom motion platforms, which can cost more than \$10,000 USD.

In addition, the analysis of display devices shows that 13% of studies still use computer screens, which represent the lowest level of immersion, as no stereoscopic effect is present and the user's surroundings are never occluded. This is a significant issue that researchers and developers must consider in subsequent studies, as it is not possible to fairly evaluate results based on screen use relative to those using the latest technology. Again, we are not advocating the inclusion of more and more feedback elements in VR systems, but basic updates, such as moving from monitors to HMDs, must be implemented. Nowadays, HMDs are affordable and can be easily purchased online.

Regarding presence assessment, two main approaches were identified. First, in subjective assessment, presence questionnaires are the most commonly used. Other tools, such as affective or cybersickness questionnaires, have been developed; however, their primary aim is not to evaluate presence and they are currently used within affective computing or as complements to presence questionnaires. These subjective tools are fundamental for assessing presence in virtual reality systems and have been the key approach since the mid-1990s. Nevertheless, their use presents several well-known limitations. Self-report measures are inherently subjective because they rely on test subjects' introspection and personal interpretation of their experiences. This subjectivity can introduce significant contrasts related to memory recall, expectations and previous experiences, and individual differences in cognitive processing between test subjects. In addition, questionnaire responses are typically collected after the virtual experience, which may result in inaccuracies due to retrospective assessment rather than real-time evaluation.

Moreover, the objective approach to assessing presence has gained attention in the research community, with efforts to acquire and process physiological signals to interpret discrete or continuous affective states. The literature review indicates a strong tendency to use ECG and EDA signals for affect detection, followed by EEG and body-tracking data. Such signals have already yielded positive results; however, other modalities, such as electromyography (EMG), should be considered when users interact with virtual environments via joysticks or haptic devices. This has already been tested on dynamic difficulty adaptation (DDA) strategies for video games [171–173], demonstrating its potential for estimating muscular fatigue, stress, and cognitive load, which are reflected in affective states.

Although objective assessment methods are a promising tool for deeper exploration of human–machine interaction elements, their implementation remains low compared to questionnaires. Figure 7 presents a summary (in percentages) of the use of subjective and objective approaches, as well as their combination for presence assessment. Among the 101 studies, 57% still use questionnaires as the primary method to assess presence, whereas 40% combine them with affect detection. This is a significant issue, as the questionnaires'

drawbacks were highlighted several years ago, and one might assume that a combined strategy should now dominate. However, this does not reject or disregard the results of 57% of studies; rather, the broad empirical and objective evidence suggests that combining both methods yields more accurate results.

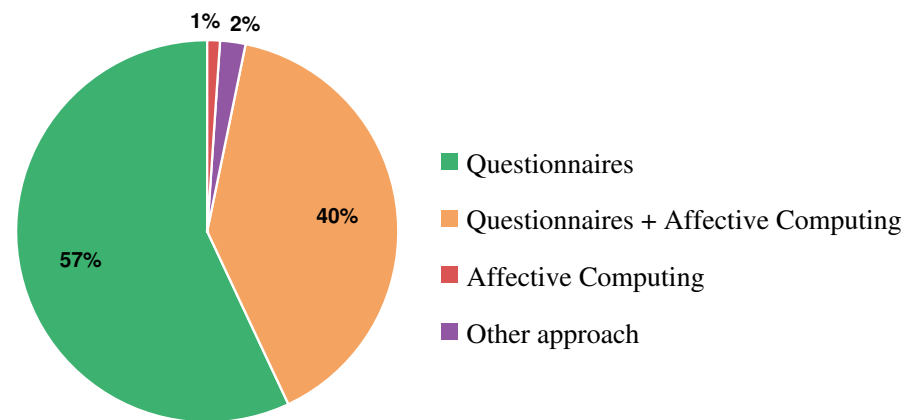


Figure 7. Percentages of presence assessment methods used in virtual reality. Questionnaires and affective computing compete with the subjective and objective methods, respectively, described in Section 4.4. Meanwhile, the category of other approaches includes alternative qualitative methodologies like semi-structured interviews [79] and qualitative affective analysis based on coded interaction indicators [111].

Beyond the chosen presence assessment method, the literature review reveals a rarely-discussed factor that could influence the replication of results and act as an indirect bias for sample populations. We discuss the representativeness of populations worldwide, which inherently reflect diverse cultural backgrounds. Figure 8 presents six continental regions around the world where the experimental evaluations of presence assessment have been carried out across the 101 reviewed articles. Most studies were conducted in Europe (despite cross-university collaboration), whereas Africa is the least represented. These statistics should be considered when new readers, researchers, and practitioners begin their research in presence assessment, so they can base their initial insights and hypotheses on similar regional results, or on completely opposite ones, to validate the universality of their methods.

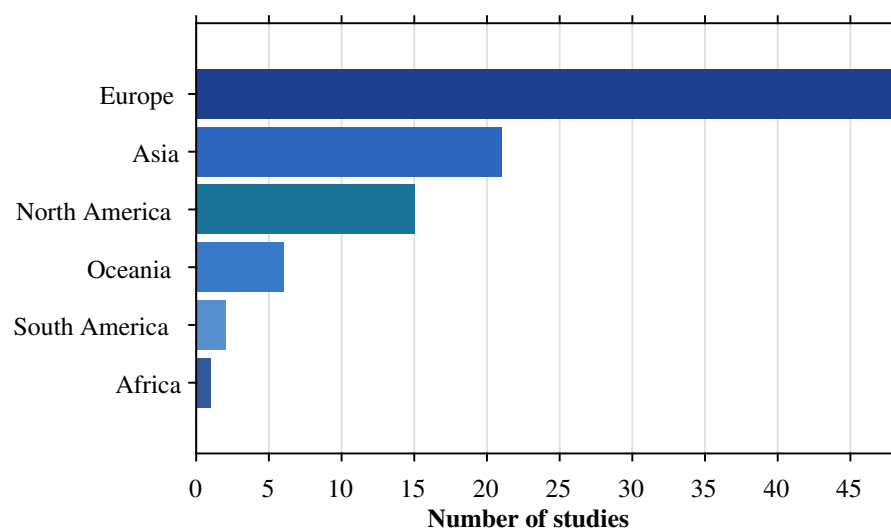


Figure 8. Worldwide regions where the experiments for assessing presence were implemented.

6. Future Perspectives

Six main paths have been identified as decisive for advancing the field. Addressing these challenges may be crucial for the continued development of more immersive and emotionally engaging VR systems, setting the stage for future innovations and applications.

1. **Develop a comprehensive and standard characterization of stimuli:** There is a need to systematically characterize a variety of stimuli of different modalities (visual, haptic, etc.), thereby facilitating the examination of a broader spectrum of affective states. Upon reviewing the existing literature, it becomes evident that the majority of experiments addressing this matter predominantly elicit states such as stress, anxiety, and heightened levels of arousal. Few studies delve into alternative emotional states or draw conclusions regarding valence and dominance. Moreover, no standard procedure currently exists to characterize and validate VR stimuli. This complicates following an overall procedure to reproduce previous literature experimental setups. For example, height exposure scenarios provide only general details of the experience without specifying the exposure heights, time, and illumination conditions in their virtual environments. These elements can vary in each setup, leading to different physiological/affective responses and presence states. Developing a VR stimuli database in popular game engines would mean a step towards achieving consistent and reproducible results.
2. **Expand the understanding of the relationship between stimuli and affective states:** Section 4.4.2 discusses the use of physiological signals to estimate discrete affective states or continuous variables like arousal and valence. The extensive empirical results and conclusions around affect detection (beyond presence assessment) do not sufficiently address the fact that the relation between stimuli and affective states is not bijective. This means that different stimuli/virtual environments can produce the same affective response. For instance, height exposure scenarios [17] and first person shooter video games [173] have been used to elicit stress, despite the different nature of the stimuli and experimental purposes. Moreover, since such relation is not bijective, there does not exist an inverse function that relates arousal/valence levels or a discrete state with an specific set of stimuli. Given this issue, the relation between each stimuli or scenario and the affective responses must be carefully analyzed and validated. This could lead to the identification of specific patterns, or a combination of them that facilitates the understanding of such a complex relationship, especially for researchers outside the field of psychology. This problem also can be advanced with the characterization of stimuli described before.
3. **Enhance the modeling strategies used in affect detection:** Most of the mathematical models involved in presence assessment consist of linear regressions or pattern recognition models. Typically, these models only determine whether presence was effectively elicited, discerning the efficacy of stimuli. Shifting this paradigm and proposing methodologies that address the question, “In which degree was presence attained by users?” will mean a step forward to advance the current results. In addition, mathematical tools like fuzzy logic, hybrid systems, or timed continuous Petri nets can be introduced to model and quantify presence, leading to its formal verification. However, the high variability among users’ characteristics, stimuli, and physiological responses makes this a challenging task that also needs to be supported by statistical tests that validate the obtained results for specific subject populations.
4. **Test closed-loop frameworks to adapt online the VR stimuli:** Most of the reviewed literature that includes experimental results works within an open-loop framework (see tenth column in Table 2). This means that participants are exposed to a VR experience with a predefined set of stimuli, physiological responses are recorded and/or

questionnaires are administered (at the end of the experience), statistical analyses are carried out, and subsequently conclusions are derived. This implies that it is not possible to modify or adjust the VR stimuli in the event of a negative VR experience. For instance, a user could face a highly demanding and stressful task or the opposite; the VR experience could become boring and unrealistic, resulting in failure to achieve presence. To change the current approach, closed-loop strategies for online control, i.e., during the experience, must complement the modeling strategies mentioned in the previous point. Some initial strategies for online-adapted VR experiences have been proposed [120,174], and others for a variety of video games [171,172,175]. However, successful implementation of the online approach faces several challenges:

- **Sample size:** More significant sample populations to verify their feasibility among subjects are required. Current literature results (e.g., [120,174]) rely on limited subject samples. Moreover, other factors, such as age intervals, gender, and prior VR experience, must be considered.
 - **textbfVariability among subjects:** This particular matter becomes challenging when dealing with physiological signals. Since each subject has specific physiological features, variability in signal responses poses a major challenge for generalizing models and analyses. Each subject has a personal baseline heart rate or skin conductivity, and, additionally, factors such as hydration, body hair, skin care products, and weather conditions can affect signal acquisition and, consequently, the results. To overcome this issue, researchers may adopt adaptive classification strategies as a basis for signal processing [176,177].
 - **Online physiological signal processing:** An important matter to take into account is that for online presence assessment and regulation of VR stimuli, physiological signals (and their characteristics) for affect detection must be evaluated in short time windows, making some signals, like EEG, not suitable for this task. Although portable systems have been developed for online signal acquisition [178,179], the most promising approach may be the multimodal fusion systems for affect detection [180].
5. **Cross-cultural research:** To address the bias introduced by cultural context among populations, it is necessary to carry out cross-cultural studies. This may involve populations of exchange students, foreign collaborators, or visiting collaborators to investigate the extent to which the designed virtual environment stimuli, tasks, or experimental setup affect their performance and presence in VR [109]. Cross-cultural studies have revealed interesting differences and affinities across populations from different regions. For instance, Šašinková et al. [181] observe differences in cognitive styles among five cultural groups exposed to visual stimuli in virtual reality, using eye-tracking technology for this purpose. Moreover, Lin et al. [182] implement an emergency evacuation scenario across three populations in different countries and observe similar behavioral patterns across evacuation tasks. Despite these examples results present valuable information among three or five populations, a remarkable path to follow by researchers will be to carry out a multi-lab, multi-region/country experiment to test VR stimuli among several populations like the experiment implemented by Vaidis et al. [183]. This will would provide comprehensive insights into the effects of cultural background, language, and other factors on users' presence, and allow testing and validation of the aforementioned modeling and closed-loop control approaches.
 6. **Presence and VR for industrial applications.** Currently the study of presence assessment has been mostly focused on testing virtual reality stimuli, signal processing methods or affective elicitation scenarios. However, its true potential can bring

benefits for industrial developments or scenarios. Is it not casually that among the main enabling technologies of Industry 4.0 there are virtual reality and digital twins (DTs) [184]. Such technologies enable the implementation of complex scenarios for training, teaching, or remote operation of machinery and processes. For instance, Pérez et al. [185] presented an automation monitoring system based on the digital twin of a manufacturing process. The digital twin is developed within an immersive virtual environment, where user experience plays a key role for human–robot collaboration in classification, assembly, inspection, and delivery of parts. Researchers should start targeting real-world applications for testing their presence assessment approaches, since real world scenarios can obtain direct benefits from virtual environments that provide plausible and engaging experiences. More emphasis should be placed on technological development beyond academic purposes.

7. Conclusions

The presence-related topics presented in Section 4 provide a broad and detailed overview of the main trends around presence in VR. In summary, 101 studies, including the keywords virtual reality, presence, sense, and assessment in the title, abstract, or keywords within a year span of 2018–2025, were analyzed. This literature review allows us to answer the four stated research questions. The most popular approach to assessing presence remains questionnaires; nonetheless, the adoption of affective computing has gained popularity as a complementary method. For affective computing purposes, the most popular physiological signals are ECG and EDA, both of which are employed to estimate continuous and discrete affective markers. Further, the majority of researchers implement virtual environments within visual and auditory cues using HMDs. Without doubt, there are still opportunities for improvement in research on VR and presence, including enhancing the sample size and considering cross-cultural factors to increase the universality of results. The implementation of closed-loop strategies is also desirable; however, it faces several challenges associated with online physiological signal processing and variability among subjects.

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Abbreviations

The following abbreviations are used in this manuscript:

AC	Affective Computing
ANS	Autonomic Nervous System
AQ	Acrophobia Questionnaire
BSIQ	Bouchard’s Single-Item Questionnaire
CIS	Checklist Individual Strength
DASS-21	Depression Anxiety Stress Scales
DEQ	Discrete Emotions Questionnaire
DTs	Digital Twins
ERQ	Emotion Regulation Questionnaire
FAS	Flight Anxiety Situations Questionnaire

FMS	Fast Motion Sickness Scale
HADS	Hospital Anxiety and Depression Scale
IPQ	Igroup Presence Questionnaire
ITC-SOPI	ITC Sense of Presence Inventory
KEDS	Kids' Empathic Development Scale
M-DAS	Modified Differential Affect Scale
MEC-SPQ	MEC Spatial Presence Questionnaire
MEDEQ	Meditation Depth Questionnaire
NMSPI	Networked Minds Measure of Social Presence
PANAS	Positive and Negative Affect Schedule
PENS-PI	PENS Presence–Immersion Subscale
PRPSA	Personal Report of Public Speaking Anxiety Scale
PSS	Perceived Stress Scale
SAM	Self-Assessment Manikin
SIP	Single Item Presence
SPES	Spatial Presence Experience Scale
SPIE	Spatial Presence in Immersive Environments Questionnaire
SSAI	Short State Anxiety Inventory
SSQ	Simulator Sickness Questionnaire
STAI	State–Trait Anxiety Inventory
SUS	Slater–Usoh–Steed Presence Questionnaire
TCI	Temperament and Character Inventory
TEIQUE-SF	Trait Emotional Intelligence Questionnaire Short Form
TPI	Temple Presence Inventory
TPI-SR	Temple Presence Inventory Social Richness
VAS	Visual Analogue Scale for Anxiety
VEQ	Virtual Embodiment Questionnaire
VHPQ	Virtual Human Plausibility Questionnaire
VRSQ	Virtual Reality Sickness Questionnaire

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